

SELECTING THE RIGHT SENSORS For Industrial Applications

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There is a very wide variety of industrial sensors available in the market, and selecting the right sensor for an application can be challenging

Industrial sensors cover almost all the sensor categories including proximity, position, velocity, level, temperature, force and pressure. These find application in extremely diverse areas including industrial process control, energy, aviation, safety and security, automobiles, healthcare and building automation to name a few.

Sometimes sensors form the core elements of products and solutions. These are an important part of the products that define quality, efficiency and safety of the intended application. From bare semiconductor chips and transmitters to complex systems, sensor solutions and technologies exist across the entire value chain. This article covers characteristics of sensors used in industrial process control and monitoring, including wireless sensors, followed by introduction to some leading manufacturers of industrial sensors.

Industrial process control

Industrial process control involves monitoring and control of machinery, systems and processes across a large number of industries including chemicals, pharmaceuticals, biotechnology, energy, water, oil and gas, plastic, paper, and food and beverages. The modern industrial processing and manufacturing systems are highly automated, ensuring that raw materials and energy are

consumed in an efficient manner. There are many parameters to be monitored and controlled including pressure, temperature, level, flow rate, humidity, dust, corrosives, explosives, liquids and gases.

Sensors are the first element in a process control and measurement system. Precise control and accurate displacement/position measurement in extremely small scales such as nanometres and picometres has become increasingly significant during the last few years. The main performance criteria for industrial sensors are sensitivity, resolution, compactness, long-term stability, thermal drift and power efficiency.

Some of the sensors used in industrial process control

Proximity sensors. These can be mechanical, optical, inductive and capacitive. They are widely used in industrial automation like conveyor lines for counting and jam detection, machine tools for safety interlock and sequencing. They are also used in detecting the presence or absence of objects.

Mechanical sensors are basically mechanical switches for on/off operations.

Optical sensors could be light sources like light-emitting diodes (LEDs) and phototransistors. These sensors are essentially non-contact type with no moving parts. They are small, fast-switching, and insensitive to vibration and shock. However, they require alignment, can be blinded by ambient light conditions, and may require clean, dust- and water-free environment.

Inductive and capacitive proximity sensors generally have a short range but are very robust and reliable. Inductive sensors use magnetic properties to detect the presence of metal objects. Capacitive sensors are normally used to detect the capacitance caused by non-metallic objects.

Position sensors. These include encoders and linear variable differential transformers

Fig. 1: Industrial process control and automation (Image courtesy: <https://www.halvorsen.blog>)

